

Week 1, Day 1 — student handout

Umesh Indrajith · Keep this sheet. SMS starts week 4.

TODAY'S OUTCOMES

- Explain software engineering in two sentences (requirements → design → implement → test → review).
- Say why we use Go for a backend API course.
- Go 1.25 + VS Code + gofmt on save.
- Hello, Epic Learn on GitHub.

INSTALL (UBUNTU). NOT APT.

Filename must match go.dev/dl on the day (1.25.0, 1.25.1, ...). Windows: do this **inside WSL Ubuntu**, not PowerShell.

```
wget https://go.dev/dl/go1.25.0.linux-amd64.tar.gz
sudo rm -rf /usr/local/go
sudo tar -C /usr/local -xzf go1.25.0.linux-amd64.tar.gz
echo 'export PATH=$PATH:/usr/local/go/bin' >> ~/.bashrc
source ~/.bashrc
go version
```

FOLDER + FIRST PROGRAM

```
mkdir -p ~/epic-go/hello
cd ~/epic-go/hello
```

VS Code: File → Open Folder → ~/epic-go/hello. Extension: **Go** (Go Team at Google).
Command Palette → **Go: Install/Update Tools**. Format on save: User Settings JSON:

```
{
  "editor.formatOnSave": true,
  "[go]": {
    "editor.defaultFormatter": "golang.go",
    "editor.formatOnSave": true
  }
}
```

```
package main

import "fmt"

func main() {
```

```
fmt.Println("Hello, Epic Learn")
}
```

Save as `hello.go`. **Println** — capital P (exported name). Then:

```
gofmt -w hello.go
go run hello.go
go build -o hello hello.go
./hello
```

Do **not** `go build .` today (no `go.mod` until week 4).

GIT → GITHUB

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"

# .gitignore – one line: hello
echo hello > .gitignore

git init
git add hello.go .gitignore
git status
git commit -m "Add Hello Epic Learn"
git branch -M main
git remote add origin https://github.com/YOU/hello-epic-learn.git
git push -u origin main
```

On github.com create an **empty** repo (no README). Then show the instructor the URL.

Clone (copy a repo — you will use this for SMS in week 4):

```
git clone https://github.com/YOU/hello-epic-learn.git
cd hello-epic-learn
```

WORDS TO REMEMBER

Term	Meaning today
GOROOT	Where Go is installed
GOPATH	Old workspace — we do not use it for class projects
Module	A folder with <code>go.mod</code> — week 4
gofmt	Official formatter; unformatted work is returned

IF IT BREAKS

- command not found → new terminal, or `source ~/.bashrc`
- Old go version → `uninstall apt golang`, use `go.dev`
- `fmt.println` → `Println`
- Push 403 → GitHub token / empty remote
- Huge file on GitHub → you committed the binary; see `.gitignore`

Exported	Name starts with a capital letter
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HOMEWORK (SAME REPO)

1. why-go.md — half page, your words (template from the instructor).
2. Proof of go version (screenshot or paste).